

IPv4 and IPv6 — Guided Notes ANSWER KEY

Unit 1.4.2 | CompTIA Network+ N10-009 Obj. 1.4 | N10-008 1.4

For instructor use / distribute after students complete the notes.

Question 1

An IPv4 address is _____ bits long, written as _____ octets in dotted decimal notation. Each octet ranges from _____ to _____. The total number of possible IPv4 addresses is approximately _____.

Answer: 32 bits; four; 0; 255; 4.3 billion.

Question 2

Every IPv4 address has two parts: a _____ portion and a _____ portion. The _____ determines where the split falls. A /24 mask means _____ bits are network and _____ bits are host.

Answer: Network; host; subnet mask; 24 bits; 8 bits.

Question 3

CIDR notation /30 leaves _____ bits for hosts, providing _____ total addresses and _____ usable addresses. This is typically used for _____.

Answer: 2 bits; 4 total addresses; 2 usable; point-to-point links between two routers.

Question 4

IPv6 addresses are _____ bits long, written as _____ groups of _____ hexadecimal digits separated by colons. Two shorthand rules are: (1) _____ and (2) _____.

Answer: 128 bits; eight groups; four hex digits; leading zeros in any group can be dropped; one consecutive span of all-zero groups can be replaced with :: (double colon, used only once per address).

Question 5

A standard IPv6 subnet assignment uses a _____ prefix, giving _____ bits for the host identifier. Devices can generate their own host portion automatically using _____ (SLAAC) without needing a DHCP server.

Answer: /64; 64 bits; Stateless Address Autoconfiguration.

Question 6

IPv6 has NO _____ addresses. Instead it uses three communication types:

- _____ — one to one
- _____ — one to many (replaces broadcast functions)
- _____ — one to nearest

Answer: Broadcast; unicast; multicast; anycast.

Question 7

Match each IPv6 address prefix to its type:

fe80::/10 → _____

2000::/3 → _____

ff00::/8 → _____

Answer: fe80::/10 → link-local unicast; 2000::/3 → global unicast (public, internet-routable); ff00::/8 → multicast.

Question 8 — Real World Application

REAL WORLD: You run ipconfig on a Windows workstation and see both a 192.168.1.45 address and a 2001:db8::42 address on the same adapter. What is this configuration called, and what does it mean for how the machine connects to the internet?

Answer: This is dual-stack. The machine has both IPv4 and IPv6 addresses active simultaneously. The OS will prefer IPv6 for connections to destinations that support it, and fall back to IPv4 for destinations that don't. Both protocols are fully functional — it's not a misconfiguration, it's the standard transition approach while the internet completes its migration to IPv6.